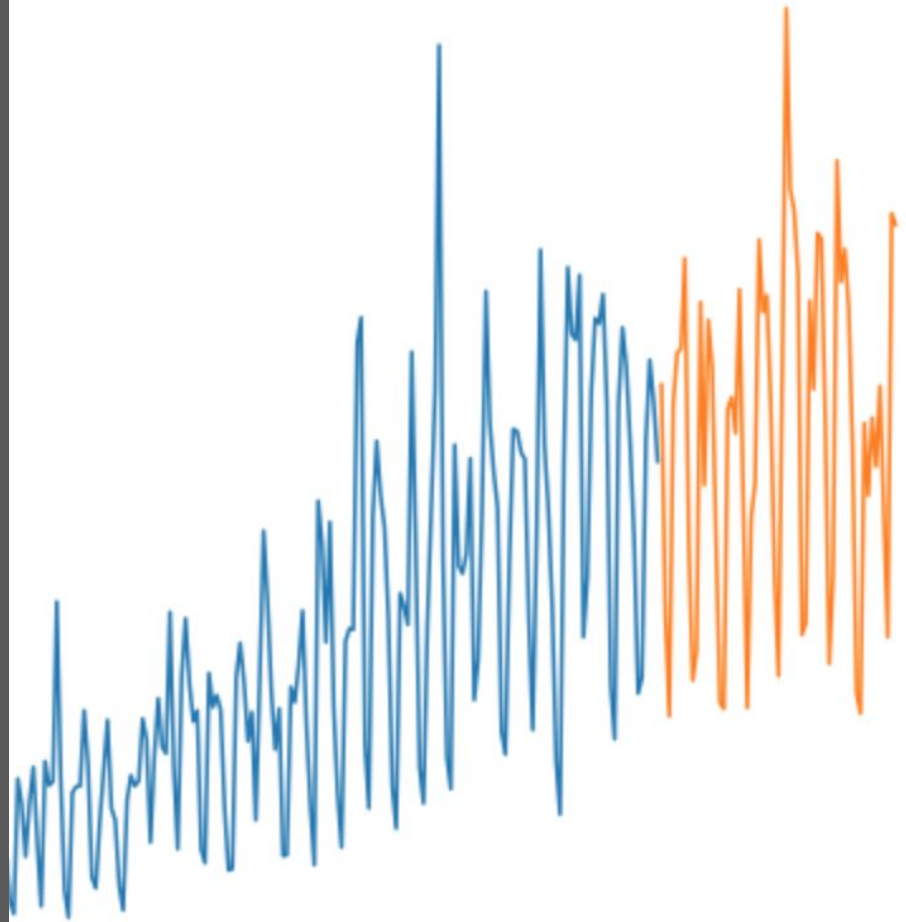


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ARIMA Modeling and Forecasting Techniques



Attendance - “ARIMA”



What is ARIMA?

- Time series forecasting model (uses time series data to predict future values from past values)
- Method built on stability and stationarity
- Has many applications:
 - Economics
 - Finance
 - Weather
 - Any scenario that includes time series data

A - Auto

R - Regression

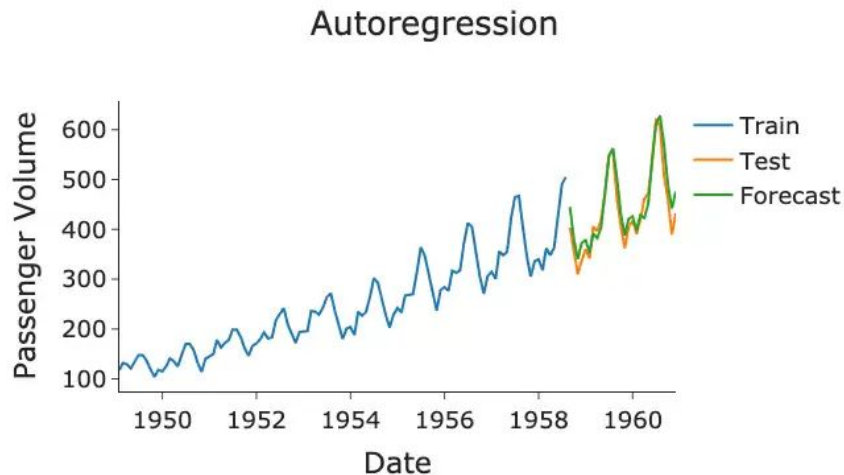
I - Integrated

M - Moving

A - Average

What is Autoregression?

- Auto = 'self'
- Instead of a normal regression model with features and a target variable, autoregression fits on the target feature's own previous lags



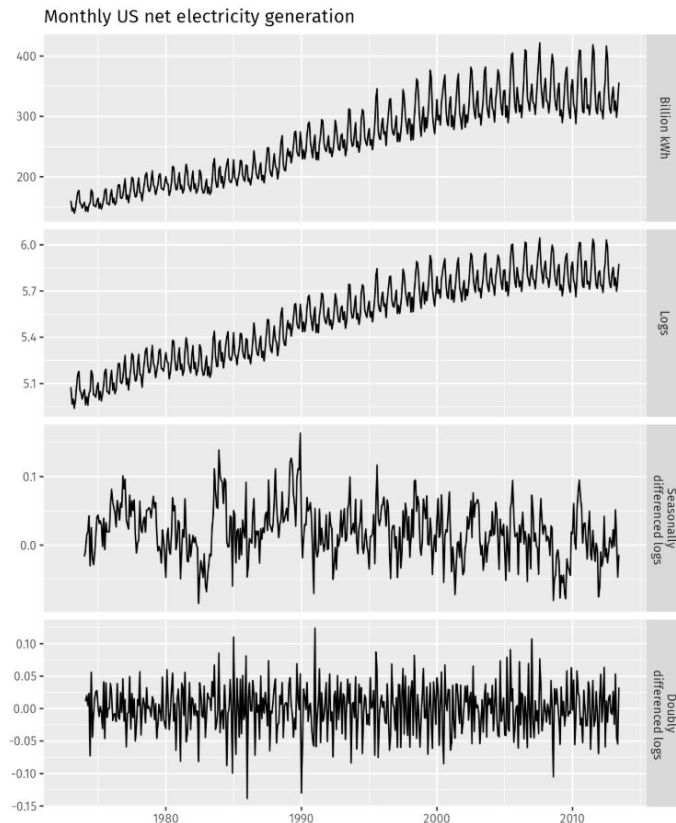
Why isn't Autoregression enough?

- Often, AR meets the forecasting needs of the data
- However, AR does not take into account the underlying statistics of the data and prediction errors
- Does not account for data that are non-stationary

$$X_t = c + \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t$$

The I: Integration by Differencing

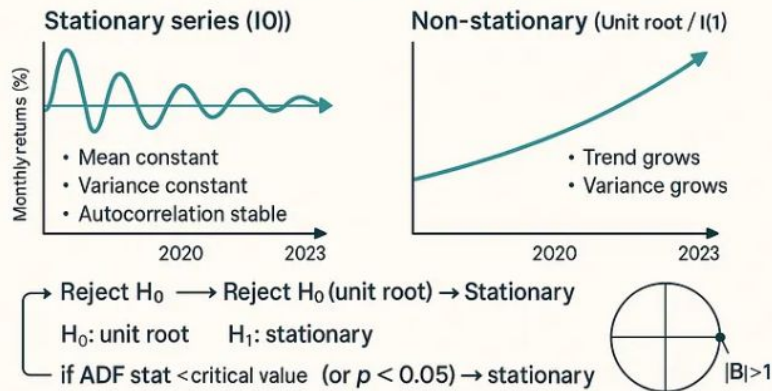
- Differencing makes non-stationary data stationary
- First-order differencing (most common)- subtract the previous observation from the current one
- Removes trends and seasonality



How to Difference the Data?

- Augmented Dickey Fuller Test shows whether the data is stationary
- Null hypothesis that the data is non-stationary
- Low p-value leads to rejecting the null hypothesis

Augmented Dickey-Fuller (ADF) Test — Quick Visual Guide



Moving Average's Role in ARIMA

- Establishes an underlying mean for past data
- Uses average error of previous forecasts to forecast future values
- Lessens volatility of forecasts (the noise)



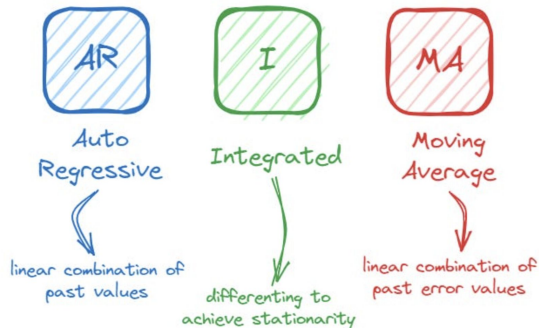
$$\hat{y}_t = \mu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} - \theta_1 e_{t-1} - \dots - \theta_q e_{t-q}$$

Putting All Components Together

- ARIMA model has three parameters: p , d , and q
 - p - number of lags to use in autoregression
 - d - how many times to difference the data
 - q - number of past error terms used in MA
- Optimizing these parameters is crucial to model success

```
# ARIMA(p, d, q) parameters
p, d, q = 2, 1, 2
model = ARIMA(train['Value'], order=(p, d, q))
model_fit = model.fit()
```

ARIMA components



Optimizing p, d, and q

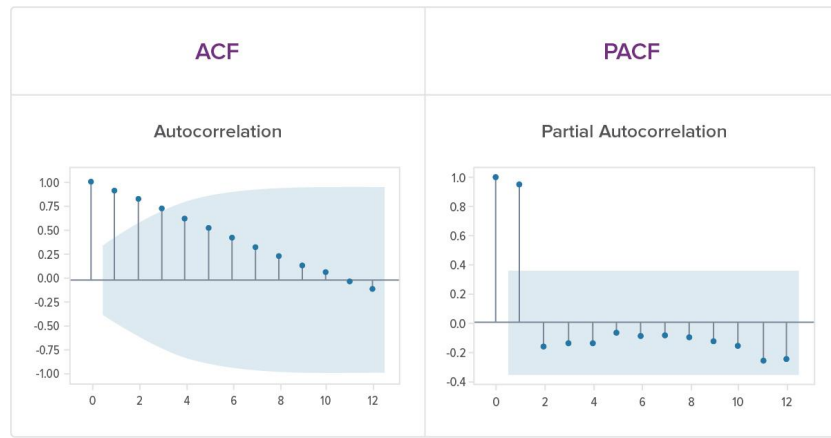
- ACF and PACF plots give good insight on which values to choose
- Sharp drop off after lag k on PACF? : $p = k$
- Use ADF test to determine stationarity for differencing
- Sharp drop off after lag k on ACF? : $q = k$

```
# ARIMA(p, d, q) parameters
```

```
p, d, q = 2, 1, 2
```

```
model = ARIMA(train['Value'], order=(p, d, q))
```

```
model_fit = model.fit()
```



Measuring Model Performance

- AIC (Akaike Information Criterion) helps compare performance of different models
- Smaller value of AIC is better
- Tune parameters to find the lowest AIC
- ar.L1 and ma.L1 show dependence on the previous data values for AR and MA

ARIMA Model Details:

SARIMAX Results

Dep. Variable:	Sales	No. Observations:	60
Model:	ARIMA(1, 1, 1)	Log Likelihood	-311.962
Date:	Wed, 30 Jul 2025	AIC	629.923
Time:	08:31:09	BIC	636.156
Sample:	01-31-2015	HQIC	632.356
	- 12-31-2019		
Covariance Type:	opg		

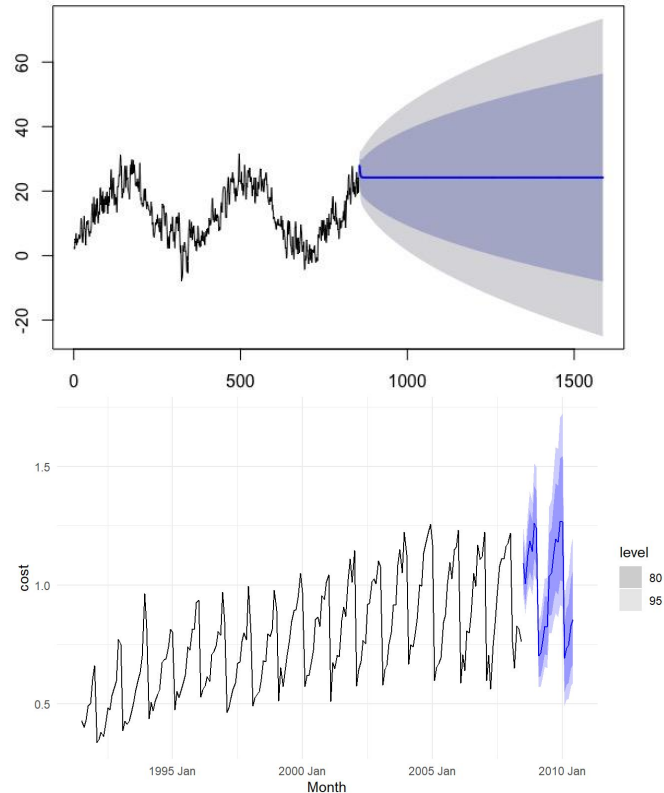
	coef	std err	z	P> z	[0.025	0.975]
ar.L1	1.0000	5.66e-05	1.77e+04	0.000	1.000	1.000
ma.L1	-0.9971	0.217	-4.600	0.000	-1.422	-0.572
sigma2	2085.9196	9.99e-05	2.09e+07	0.000	2085.919	2085.920

Ljung-Box (L1) (Q):	0.14	Jarque-Bera (JB):	0.69
Prob(Q):	0.71	Prob(JB):	0.71
Heteroskedasticity (H):	0.67	Skew:	0.01
Prob(H) (two-sided):	0.38	Kurtosis:	2.47

Walk Forward Approach

- Model can only realistically predict 'lag' steps forward
- Walk Forward Approach allows for moving predictions with new information
- Must retrain model with each new actual value and predict – can be computationally expensive

Forecasts from ARIMA(1,1,2)



Colab

